

Project Design Phase

Solution Architecture

Date	30 September 2026
Team ID	SWTID-2026-5397
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Team Members

Role	Name
Team Leader	LAVANYA S
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Team Member	LITHIKA P
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Solution Architecture

Solution architecture defines how the business requirement is translated into a technical workflow. For this project, spreadsheet data is imported into ServiceNow through an Import Set and processed using a Transform Map. The Transform Map controls field mapping, data conversion, validation, and the final loading of records into the target table.

- Provide a controlled method for loading spreadsheet data into ServiceNow.
- Map source spreadsheet columns to the correct target-table fields.
- Validate and transform incoming values before records are created or updated.
- Reduce manual data entry and improve consistency of imported records.
- Maintain an import history and error information for troubleshooting.

Architecture & Data Flow

Spreadsheet Excel / CSV data	Import Set Staging table	Transform Map Field mapping + rules	Target Table ServiceNow records
Source data prepared by team	Raw data is loaded	Transform / validate / coalesce	Validated records available

Data flow: Spreadsheet → Import Set (staging) → Transform Map → Target Table

Transform Map Processing

Step	Description
1. Prepare	Create/clean the spreadsheet with the required columns and valid values.

Step	Description
2. Load	Upload the spreadsheet into ServiceNow and create an Import Set.
3. Map	Create a Transform Map and map source columns to target-table fields.
4. Transform	Use field maps, scripts, default values, and data conversion where required.
5. Coalesce	Identify a unique field when existing records should be updated instead of duplicated.
6. Validate	Run the transform and review successful records and any errors.
7. Verify	Check the target table and confirm that imported records contain the expected values.

Implementation Details

Import Data using Transform Maps (Spreadsheet)

Key Components

Component	Purpose
Spreadsheet	Contains source records and source field names.
Import Set	Temporary staging area used to load source data.
Transform Map	Defines how staging fields are transformed and mapped to the target table.
Field Maps	Connect individual source fields to target fields.
Coalesce	Determines whether a matching record should be updated rather than inserted.
Target Table	Stores the final ServiceNow records after transformation.

Example Source Spreadsheet

Employee ID	Name	Department	Email
EMP001	Anita	IT	anita@example.com
EMP002	Kumar	HR	kumar@example.com
EMP003	Priya	Finance	priya@example.com

Expected Outcome

- Spreadsheet records are successfully loaded into the Import Set.
- Source fields are mapped to the required target fields through the Transform Map.
- Invalid or incomplete records can be identified through transform errors.
- Existing records can be updated using an appropriate coalesce field.
- The final target table contains clean and usable ServiceNow records.

Conclusion

The proposed architecture provides a simple and repeatable way to import spreadsheet data into ServiceNow. Using Import Sets and Transform Maps separates raw data loading from transformation and target-table insertion, making the process easier to validate, monitor, and maintain.

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